

CHAPTER THREE

Measure of Dispersion Or Measure of Variability

Introduction to measure of dispersion

A modern student of statistics is mainly interested in the study of variability and uncertainty. In this section we shall discuss variability and its measures and uncertainty will be discussed in probability. We live in a changing world. Changes are taking place in every sphere of life. A man of statistics does not show much interest in those things which are constant. The total area of the earth may not be very important to a research minded person but the area under different crops, area covered by forests, area covered by residential and commercial buildings are figures of great importance because these figures keep on changing from time to time and from place to place. Very large number of experts is engaged in the study of changing phenomenon. Experts working in different countries of the world keep a watch on forces which are responsible for bringing changes in the fields of human interest. The agricultural, industrial and mineral production and their transportation from one part to the other parts of the world are the matters of great interest to the economists, statisticians, and other experts. The changes in human population, the changes in standard living, and changes in literacy rate and the changes in price attract the experts to make detailed studies about them and then correlate these changes with the human life. Thus variability or variation is something connected with human life and study is very important for mankind.

Dispersion:

The word dispersion has a technical meaning in statistics. The average measures the center of the data. It is one aspect observations. Another feature of the observations is as to how the observations are spread about the center. The observation may be close to the center or they may be spread away from the center. If the observation are close to the center (usually the arithmetic mean or median), we say that dispersion, scatter or variation is small. If the observations are spread away from the center, we say dispersion is large. Suppose we have three groups of students who have obtained the following marks in a test. The arithmetic means of the three groups are also given below:

GroupA:46,48,50,52,54

$$\bar{X}_A = 50$$

GroupB:30,40,50,60,70

$$\bar{X}_B = 50$$

GroupC:40,50,60,70,80

$$\bar{X}_C = 60$$

In a group A and B arithmetic means are equal i.e. $\bar{X}_A = \bar{X}_B = 50$. But in group A the observations are concentrated on the center. All students of group A have almost the same level of performance. We say that there is consistence in the observations in group A. In group B the mean is 50 but the observations are not closed to the center. One observation is as small as 30 and one observation is as large as 70. Thus there is greater dispersion in group B. In group C the mean is 60 but the spread of the observations with respect to the center 60 is the same as the spread of the observations in group B with respect to their own center which is 50. Thus in group B and C the means are different but their dispersion is the same. In group A and C the means are different and their dispersions are also different. Dispersion is an important feature of the observations and it is measured with the help of the measures of dispersion, scatter or variation. The word variability is also used for this idea of dispersion. The study of dispersion is very important in statistical data. If in a certain factory there is consistence in the wages of workers, the workers will be satisfied. But if some workers have high wages and some have low wages, there will be unrest among the low paid workers and they might go on strikes and arrange demonstrations. If in a certain country some people are very poor and some are very high rich, we say there is economic disparity. It means that dispersion is large. The idea of dispersion is important in the study of wages of workers, prices of commodities, standard of living of different people, distribution of wealth, distribution of land among framers and various other fields of life. Some brief definitions of dispersion are:

1. The degree to which numerical data tend to spread about an average value is called the dispersion or variation of the data.
2. Dispersion or variation may be defined as a statistics signifying the extent of the scatteredness of items around a measure of central tendency.
3. Dispersion or variation is the measurement of the scatter of the size of the items of a series about the average.

Definemeasures of dispersion.

Ans:Dispersion is an important characteristic of frequency distributions. It tells how completely the individual values are distributed around the average.

According to Dr. A. L. Bowley, “Dispersion is the measure of variation of the items.”

Spiegel defined, “The degree to which the numerical data tends to spread about an average value is called dispersion or variation of data.”

Importance of measure of dispersion

The central value like mean is generally used to convey the general behavior of a data set. For example, the average score of the class in a math test hints at the general comfort level of the class in the topic which is tested. But, if two classes have the same average score, can the teacher conclude that the understanding level is same for the two classes? The arithmetic mean does not convey the variations displayed in the individual marks of the students. The teacher needs to have some idea about the spread of the marks of both the classes. Teacher needs some numerical measure of dispersion which would convey how the marks are spread about the central value, the mean in this case.

Q :

A measure of dispersion is a good supplement to the central value in understanding a frequency distribution. Comment.

Answer :

The study of the averages is only one sided distribution story. In order to understand the frequency distribution fully, it is essential to study the variability of the observations. The average measures center of the data whereas the quantum of the variation is measured by the measures of dispersion like range, quartile deviation, mean deviation and Standard Deviation. For example, if a country has very high income group people and very low income group people, then we can say that the country has large income disparity.

Types of measures dispersion.

There are two types of dispersions:

- (I) Absolute measure of dispersion
- (II) Relative measures of dispersion

Absolute measure of dispersion:

These measures give us an idea about the amount of dispersion in a set of observations. They give the answers in the same units as the units of the original observations.

Different types of absolute measures are:

- (i) Range
- (ii) **Semi-interquartile range** or Quartile deviation
- (iii) Mean deviation
- (iv) Variance and standard deviation

Relative measures of dispersion

- (i) Coefficient of Range
- (ii) Coefficient of quartile deviation

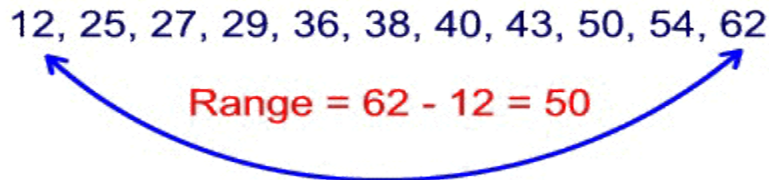
- (iii) Coefficient of mean deviation
- (iv) Coefficient of variation (CV)

Range:

Range is the difference of the highest and the lowest observations of the distributions.

Range = Maximum value- Minimum value.

Example:



For grouped data, range is the difference between the highest observation in the last class interval and the lowest observation in the first class interval. Range can never be zero.

Test-1: Find the range of the following data set:

{1; 4; 5; 8; 6; 7; 5; 6; 7; 4; 10; 9; 10}

Advantages of Range:

- (i) It is very easy to understand and easy to calculate
- (ii) It gives a quick idea about the variability of a set of data
- (iii) It is the simplest of all measures of dispersion

Disadvantages of Range:

- (i) It is very much affected by extreme values;
- (ii) It provides us with an idea of only two extreme values in a set of data;
- (iii) It cannot be computed for data set having open ended class interval

Quartile deviation/: Semi-interquartile range:

If Q_1 & Q_3 denote the first and third quartiles respectively of a frequency distribution, the quartile deviation of the data is defined by the relation -

Inter quartile range = $Q_3 - Q_1$ and

Quartile deviation/: Semi-interquartile range, $Q.D = \frac{Q_3 - Q_1}{2}$

Advantages of Quartile deviation:

- (i). It is easily understandable
- (ii) The quartile deviation is not affected by extreme values.
- (iii) It is not possible to compute-quartile deviation for open ended class

Disadvantages of Quartile deviation:

- (i). It is not based upon all the observations.
- (ii) It is more affected by sampling fluctuations
- (iii) It is not suitable for further algebraic treatment

Problem 1: Find the range of the data set

{1; 2; 3; 4; 4; 4; 5; 6; 7; 8; 8; 9; 10; 10}

Problem 2: What are the quartiles of this data set?

{3; 5; 1; 8; 9; 12; 25; 28; 24; 30; 41; 50}

Problem 3: A class of 12 student's writes a test and the results are as follows:

20; 39; 40; 43; 43; 46; 53; 58; 63; 70; 75; 91

Find the range, quartiles and the interquartile range.

Problem 4: Three sets of data are given:

- **Data set 1: {9; 12; 12; 14; 16; 22; 24}**
- **Data set 2: {7; 7; 8; 11; 13; 15; 16; 16}**
- **Data set 3: {11; 15; 16; 17; 19; 19; 22; 24; 27}**

For each data set find:

1. the range
2. the lower quartile
3. the interquartile range
4. the semi-interquartile range
5. the median
6. the upper quartile

Each set is already ordered, so we do not need to order the sets.

- a) The range is the difference between the highest and lowest values.
- b) Data set 1: The highest value is 24 and the lowest value is 9. The range is $24-9 = 15$
- c) Data set 2: The highest value is 16 and the lowest value is 7. The range is $16-7 = 9$
- d) Data set 3: The highest value is 27 and the lowest value is 11. The range is $27-11 = 16$

f) The quartiles divide the data into four equal parts. We count how many values there are in each data set and then divide this by 4. This will mark out each data set into 4 equal parts and give us the quartiles. Note that the middle quartile is the median.

Data set 1: There are 7 values in the data set. $\frac{7}{4} = 1, 75$

So the lower quartile is 12, the median is 14 and the upper quartile is 22.

he interquartile range is $Q_3 - Q_1 = 22 - 12 = 10$

The semi-interquartile range is $\frac{Q_3 - Q_1}{2} = \frac{10}{2} = 5$

Data set 2: There are 8 values in the data set. $\frac{8}{4} = 2$

In this case we need to take the average of the two numbers that lie on either side of the split to get the quartile.

So the lower quartile is $\frac{7+8}{2} = 7, 5$, the median is $\frac{11+13}{2} = 12$ and the upper quartile is $\frac{15+16}{2} = 15, 5$.

The interquartile range is $Q_3 - Q_1 = 15, 5 - 7, 5 = 8$

The semi-interquartile range is $\frac{Q_3 - Q_1}{2} = \frac{8}{2} = 4$

Data set 3: There are 9 values in the data set. $\frac{9}{4} = 2, 25$

This is slightly trickier to see how the numbers are equally divided. We will work out the median first. The median is the value at position $\frac{9+1}{2} = 5$. This is 19.

11 15 | 16 17 19 19 22 | 24 27
median

The lower quartile is $\frac{15+16}{2} = 15, 5$ and the upper quartile is $\frac{22+24}{2} = 23$.

The interquartile range is $Q_3 - Q_1 = 23 - 15, 5 = 7, 5$

The semi-interquartile range is $\frac{Q_3 - Q_1}{2} = 7, \frac{5}{2} = 3, 75$

Mean deviation:

The average deviation is the arithmetic mean of the absolute values of the deviations from the mean/median/mode.

For ungrouped data, mean deviation –

- a) about mean: $M.D.(\bar{x}) = \frac{\sum_{i=1}^n |x_i - \bar{x}|}{n}$
- b) about median: $M.D.(M_e) = \frac{\sum_{i=1}^n |x_i - M_e|}{n}$
- c) about mod: $M.D.(M_o) = \frac{\sum_{i=1}^n |x_i - M_o|}{n}$

For grouped data or frequency distributions,

- Mean deviation –about mean: $M.D.(\bar{x}) = \frac{\sum_{i=1}^n f_i |x_i - \bar{x}|}{n}$
- b) about median: $M.D.(M_e) = \frac{\sum_{i=1}^n f_i |x_i - M_e|}{n}$
- c) about mod: $M.D.(M_o) = \frac{\sum_{i=1}^n f_i |x_i - M_o|}{n}$

Learn by heart: Mean deviation

Ungroup data	Group data
$M.D.(\bar{x}) = \frac{\sum_{i=1}^n x_i - \bar{x} }{n}$	$M.D.(\bar{x}) = \frac{\sum_{i=1}^n f_i x_i - \bar{x} }{n}$

Example-1: Mean Deviation of 3, 6, 6, 7, 8, 11, 15, 16

$$\bar{x} = \frac{3 + 6 + 6 + 7 + 8 + 11 + 15 + 16}{8} = \frac{72}{8} = 9$$

x	$ x - \bar{x} $
3	6
6	3
6	3
7	2
8	1
11	2
15	6
16	7
	$\Sigma x - \bar{x} = 30$

$$\text{Mean Deviation} = \frac{\Sigma|x - \bar{x}|}{n} = \frac{30}{8} = 3.75$$

Variance:

The variance is the arithmetic mean of the squared deviations from the mean. It is denoted by σ^2

Standard deviation:

The positive square root of the variance is called standard deviation. It is denoted by σ Let,

$$x_1, x_2, x_3, \dots, x_n \text{ be the series of } n \text{ observations, then variance, } \sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

And standard deviation is

$$\begin{aligned}
 \sigma &= \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}} \\
 \Rightarrow \sigma^2 &= \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n} \\
 \Rightarrow \sigma^2 &= \frac{\sum_{i=1}^n (x_i^2 - 2x_i \cdot \bar{x} + \bar{x}^2)}{n} \\
 \Rightarrow \sigma^2 &= \frac{\sum_{i=1}^n x_i^2 - 2\bar{x} \sum_{i=1}^n x_i + n\bar{x}^2}{n} \\
 \Rightarrow \sigma^2 &= \frac{\sum_{i=1}^n x_i^2 - 2\bar{x} \cdot n\bar{x} + n\bar{x}^2}{n} \\
 \Rightarrow \sigma^2 &= \frac{\sum_{i=1}^n x_i^2 - 2n\bar{x}^2 + n\bar{x}^2}{n} \\
 \Rightarrow \sigma^2 &= \frac{\sum_{i=1}^n x_i^2 - n\bar{x}^2}{n} \\
 \Rightarrow \sigma^2 &= \frac{\sum_{i=1}^n x_i^2}{n} - \bar{x}^2 = \frac{\sum_{i=1}^n x_i^2}{n} - \left(\frac{\sum_{i=1}^n x_i}{n}\right)^2 \\
 \Rightarrow \sigma &= \sqrt{\frac{\sum_{i=1}^n x_i^2}{n} - \left(\frac{\sum_{i=1}^n x_i}{n}\right)^2} \dots\dots\dots(1)
 \end{aligned}$$

$$\text{Here, } \bar{x} = \frac{\sum_{i=1}^n x_i}{n} \Rightarrow n\bar{x} = \sum_{i=1}^n x_i$$

Equation (1) is called the working formula of standard deviation for ungrouped data.

Again, for grouped data or frequency distributions, variance is –

$$\sigma^2 = \frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{n} .$$

∴

Standard deviation is –

$$\sigma = \sqrt{\frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{n}} \quad \text{Now,}$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{n}}$$

$$\Rightarrow \sigma^2 = \frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{n}$$

$$\Rightarrow \sigma^2 = \frac{\sum_{i=1}^n f_i (x_i^2 - 2x_i \bar{x} + \bar{x}^2)}{n}$$

$$\Rightarrow \sigma^2 = \frac{\sum_{i=1}^n f_i x_i^2 - 2\bar{x} \sum_{i=1}^n f_i x_i + \sum_{i=1}^n f_i \bar{x}^2}{n}$$

$$\Rightarrow \sigma^2 = \frac{\sum_{i=1}^n f_i x_i^2 - 2\bar{x} \cdot n\bar{x} + n\bar{x}^2}{n}$$

$$\Rightarrow \sigma^2 = \frac{\sum_{i=1}^n f_i x_i^2 - 2n\bar{x}^2 + n\bar{x}^2}{n}$$

$$\Rightarrow \sigma^2 = \frac{\sum_{i=1}^n f_i x_i^2 - n\bar{x}^2}{n}$$

$$\Rightarrow \sigma^2 = \frac{\sum_{i=1}^n f_i x_i^2}{n} - \bar{x}^2 = \frac{\sum_{i=1}^n f_i x_i^2}{n} - \left(\frac{\sum_{i=1}^n f_i x_i}{n} \right)^2$$

$$\Rightarrow \sigma = \sqrt{\frac{\sum_{i=1}^n f_i x_i^2}{n} - \left(\frac{\sum_{i=1}^n f_i x_i}{n} \right)^2} \dots\dots\dots(2)$$

$$\text{Here, } \bar{x} = \frac{\sum_{i=1}^n f_i x_i}{n} \Rightarrow n\bar{x} = \sum_{i=1}^n f_i x_i$$

Equation (2) is called the working formula for frequency distribution of grouped data.

Learn by heart: Variance

Ungroup data	Group data
$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$	$\sigma^2 = \frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{n}$

Learn by heart:Standard deviation

Ungroup data	Group data
$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$	$\sigma = \sqrt{\frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{n}}$

Table 1. Comparison of Sample Statistics and Population Parameters

	Sample Statistic	Population Parameter
Mean	$\bar{x}$	μ
Standard deviation	s	sigma
Variance	s^2	sigma ²

Example 1 – Variance and standard deviation (For ungroup data):

****Find variance and standard deviation for the following data set.**

5, 10, 8, 12,20,24,25,15,16,22

Solution: we know,

$$\begin{aligned}\bar{X} &= \frac{\sum_{i=1}^n x_i}{n} \\ &= \frac{5+10+8+12+20+24+25+15+16+22}{10} = \frac{157}{10} = 15.7\end{aligned}$$

Table for calculating variance and standard deviation:

X_i	$(X_i - \bar{X})$ $\bar{X} = 15.7$	$(X_i - \bar{X})^2$
5	-10.7	114.49
10	-5.7	32.49
8	-7.7	59.29
12	-3.7	13.69
20	4.3	18.49
24	8.3	68.89
25	9.3	86.49
15	-0.7	0.49
16	0.3	0.09
22	6.3	36.69
$\sum_{i=1}^n x_i = 157$		$\sum_{i=1}^{10} (x_i - \bar{x})^2 = 434.1$

We know, variance

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$$= \frac{434.1}{10} = 43.41$$

And standard deviation

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$$

$$\sigma = \sqrt{\frac{434.1}{10}} = \sqrt{43.41} = 6.59$$

Alternative method of Variance and standard deviation

We know, Variance, $\sigma^2 = \frac{\sum_{i=1}^n x_i^2}{n} - \left(\frac{\sum_{i=1}^n x_i}{n} \right)^2$

X_i	x_i^2
5	25
10	100
8	64
12	144
20	400
24	576
25	625
15	225
16	256
22	484
$\sum_{i=1}^n x_i = 157$	$\sum_{i=1}^n x_i^2 = 2899$

Therefore, $\sigma^2 = \frac{\sum_{i=1}^n x_i^2}{n} - \left(\frac{\sum_{i=1}^n x_i}{n} \right)^2 = \frac{2899}{10} - \left(\frac{157}{10} \right)^2 = 289.9 - 246.49 = 43.41$

And standard deviation

$$\sigma = \sqrt{43.41}$$

$$= 6.59$$

Example 2 – Standard deviation(For ungroup data)

A hen lays eight eggs. Each egg was weighed and recorded as follows:

60 g, 56 g, 61 g, 68 g, 51 g, 53 g, 69 g, 54 g.

1. First, calculate the mean:

$$\begin{aligned}\bar{x} &= \frac{\sum x}{n} \\ &= \frac{472}{8} \\ &= 59\end{aligned}$$

2. Now, find the standard deviation.

Table 1. Weight of eggs, in grams		
Weight (x)	(x - $\bar{x}$)	(x - $\bar{x}$) ²
60	1	1
56	-3	9
61	2	4
68	9	81
51	-8	64
53	-6	36
69	10	100
54	-5	25
472		320

3. Using the information from the above table, we can see that

$$\sum (x - \bar{x})^2 = 320$$

In order to calculate the standard deviation, we must use the following formula:

$$\begin{aligned}s &= \sqrt{\frac{\sum (x - \bar{x})^2}{n}} \\ &= \sqrt{\frac{320}{8}} \\ &= 6.32 \text{ grams}\end{aligned}$$

Exercises 1–6, find Mean deviation, variance and standard deviation of the following set of measurements.

- (1) 5, 12, 7, 14, 8, 9, 7
- (2) 30, 37, 32, 39, 33, 34, 32
- (3) 5, 12, 7, 24, 8, 9, 7
- (4) 20, 37, 32, 39, 33, 34, 32
- (5) 5, 12, 7, 14, 9, 7
- (6) 30, 37, 32, 39, 34, 32

Variance and standard deviation (For group data):

Example-1: Find variance and standard deviation for the following data set.

Class interval	5-10	10-15	15-20	20-25
Frequency	2	5	8	3

Solution: Table for calculating variance and standard deviation:

Class interval	Frequency (f)	Mid value (x _i)	$f_i x_i$	x_i^2	$f_i x_i^2$
5-10	2	7.5	15	56.25	112.5
10-15	5	12.5	62.5	156.25	781.25
15-20	8	17.5	140	306.25	2450
20-25	3	22.5	67.5	506.25	1518.75
			$\sum_{i=1}^n f_i x_i = 285$		$\sum_{i=1}^n f_i x_i^2 = 4862.5$

$$\begin{aligned} \text{We know, Variance, } \sigma^2 &= \frac{\sum_{i=1}^n f_i x_i^2}{n} - \left(\frac{\sum_{i=1}^n f_i x_i}{n} \right)^2 \\ &= \frac{4862.5}{18} - \left(\frac{285}{18} \right)^2 = 270.89 - (15.8)^2 = 270.1389 - 249.64 \\ &= 20.49889 \end{aligned}$$

And standard deviation

$$\begin{aligned} \sigma &= \sqrt{20.49889} \\ &= 4.53 \end{aligned}$$

Exercise-1: Find variance and standard deviation for the following data set.

Class interval	10-20	20-30	30-40	40-50	50-60
Frequency	5	8	9	5	3

Properties of standard deviation

1. The standard deviation is zero if all the observations under study are same
2. Standard deviation is independent of change of origin but not of scale
3. For two observations, standard deviation is the half of the range $\left| \frac{x_1 - x_2}{2} \right|$
4. It is suitable for further algebraic treatment
5. Standard deviation of n natural numbers is $\sqrt{\frac{n^2 - 1}{12}}$

If $\bar{x}$ = mean, S = standard deviation and x = a value in the data set, then

- About 68% of the data lie in the interval: $\bar{x} - S < x < \bar{x} + S$.
- About 95% of the data lie in the interval: $\bar{x} - 2S < x < \bar{x} + 2S$.
- About 99% of the data lie in the interval: $\bar{x} - 3S < x < \bar{x} + 3S$.

Relative measure of dispersion:

These measures are calculated for the comparison of dispersion in two or more than two sets of observations. These measures are free of the units in which the original data is measured. These measures are a sort of ratio and are called coefficients.

Different types of relative measures are:

i) Co efficient of range:

The co-efficient of range is defined as $C.R. = \frac{x_{\max} - x_{\min}}{x_{\max} + x_{\min}} \times 100$ where, $x_{\max}$ and $x_{\min}$ is the highest and lowest observation respectively

ii) Co-efficient of quartile deviation:

The co-efficient of quartile deviation is defined as $C.Q.D. = \frac{Q_3 - Q_1}{Q_3 + Q_1} \times 100$, where Q_3 and Q_1 is the third and first quartile respectively.

iii) Co-efficient of mean deviation:

The co-efficient of mean deviations are:

- a) about mean, defined as $C.M.D.(\bar{x}) = \frac{M.D.(\bar{x})}{\bar{x}} \times 100$

- b) about median, defined as $C.M.D.(M_e) = \frac{M.D.(M_e)}{M_e} \times 100$
- c) about mod, as defined as $C.M.D.(M_o) = \frac{M.D.(M_o)}{M_o} \times 100$

Learn by heart:iv) Co-efficient of variation:

The **coefficient of variation** is the **ratio** between the **standard deviation** of a sample and its **mean**.

$$\text{i.e, } C.V = \frac{\sigma}{x}$$

The coefficient of variation is usually expressed in percentages:

$$C.V = \frac{\sigma}{x} \times 100$$

It allows us to compare the dispersions of two different distributions if their means are positive.

Example to explain the coefficient of variation

Example-1: A distribution is $\bar{x} = 140$ and $\sigma = 28.28$ and the other is $\bar{x} = 150$ and $\sigma = 24$

Which of the two has a greater dispersion?

Solution:

$$C.V_1 = \frac{28.28}{140} \cdot 100 = 20.2\%$$

$$C.V_2 = \frac{24}{150} \cdot 100 = 16\%$$

The first distribution has a higher dispersion.

Example-2: Suppose the mean height of 200 students was 150 Cm with a standard deviation of 12 cm and mean weight was 60 kg with a standard deviation of 7 kg.

Then Coefficient of variation of height = $\frac{12}{150} \times 100 = 8\%$

Coefficient of variation of weight = $\frac{7}{60} \times 100 = 11.67\%$

Thus weight has a higher variability than the height. The coefficient of variation is used to compare the variability, homogeneity, stability, uniformity and consistency of two series. Smaller the coefficient of variation, higher is the consistency, homogeneity, stability, uniformity etc.

Example-3: If the laboratory technician A completes 40 analyses with a standard deviation of 5 and technician B completes 160 analyses per day with standard deviation of 15, find which employee shows less variability.

Solution: Compute the coefficient of variation for both technicians

C. V for technician A = $\frac{5}{40} \times 100 = 12.5 \%$

C. V for technician B = $\frac{15}{60} \times 100 = 9.4 \%$

So, we find that technician B less variation than technician A.

Uses of Co-efficient of variation:

Without an understanding of the relative size of the standard deviation compared to the original data, the standard deviation is somewhat meaningless for use with the comparison of data sets. To address this problem the coefficient of variation is used.

Problem-1: Two cricketers scored the following runs in randomly selected 10 one day matches:

Player-A	42	32	40	45	17	83	59	64	76	72
Player-B	95	3	28	70	31	14	82	0	59	108

(i) Which player has higher average run?

(ii) Which player is more consistent?

Solution: Table for calculating coefficient of variation:

Player-A (x_i)	Player-B (y_i)	x_i^2	y_i^2
42	95	1764	9025
32	3	1024	9
40	28	1600	784
45	70	2025	4900
17	31	289	961
83	14	6889	196
59	82	3481	6724
64	0	4096	0
76	59	5776	3481
72	108	5184	11664
$\sum_{i=1}^n x_i = 530$	$\sum_{i=1}^n y_i = 490$	$\sum_{i=1}^n x_i^2 = 32127$	$\sum_{i=1}^n y_i^2 = 37744$

$$(i) \quad \text{Player-A: } \overline{X}_A = \frac{\sum_{i=1}^n x_i}{n}$$

$$= \frac{530}{10} = 53$$

$$\text{Player: } \overline{y}_B = \frac{\sum_{i=1}^n y_i}{n}$$

$$= \frac{490}{10} = 49$$

Since the average score for the player A is higher than player B, hence A has higher average run.

(ii) We know that,

$$C.V = \frac{\sigma}{\overline{x}} \times 100 \text{ -----(i)}$$

$$\sigma_A^2 = \frac{\sum_{i=1}^n x_i^2}{n} - (\overline{x})^2$$

$$= \frac{32127}{10} - (53)^2 = 3212.8 - 2809 = 403.7$$

Therefore $\sigma_A = 20.09$

$$C.V (A) = \frac{\sigma_A}{\overline{x}_A} \times 100 = \frac{20.09}{53} \times 100 = 37.90\%$$

Similarly,

$$C.V = \frac{\sigma}{\overline{x}} \times 100 \text{ -----(ii)}$$

$$\sigma_B^2 = \frac{\sum_{i=1}^n y_i^2}{n} - (\overline{y})^2$$

$$= \frac{37744}{10} - (49)^2 = 3774.4 - 2401 = 1373.4$$

Therefore $\sigma_B = 37.05$

And C.V (B) = $\frac{\sigma_B}{x_B} \times 100 = \frac{37.05}{49} \times 100 = 75.61\%$

The coefficient variation for player A is less than player B; hence player A is more consistent.

(iii) Player A will get the prize

Similar problem-1: Run of two batsmen of last 5 ODI cricket matches is given below:

Batsman A	65	68	25	78	85
Batsman B	40	100	35	18	28

(i) Which batsman has higher average run?

(ii) Which batsman is more consistent?

Similar problem-2: CGPA of two sections of students, each section containing 8 students, which are given below:

CGPA of Section A	3.90	4.00	3.75	3.50	2.90	3.45	3.80	3.95
CGPA of Section B	4.00	2.95	3.50	3.25	3.59	3.80	3.60	3.90

i) Which section students CGPA is more than other section?

ii) In which section students results is more consistent than other section?

Similar problem-3: Daily wages of workers of two factories A& B are as follows:

Daily wages (Tk.)	90-99	100-109	110-119	120-129	130-139
No. of workers of factory A	6	13	20	6	5
No. of workers of factory B	4	8	29	4	5

i) Which factory has less variability wages?

ii) Find combined mean and standard deviation of wages of the factories.

Theorem: For n non-negative observations, show that, $CV \leq 100\sqrt{n-1}$ or $\bar{x}\sqrt{n-1} \geq \sigma$.

Proof: Let, $x_1, x_2, x_3, \dots, x_n$ be the set of n observation. Then, their arithmetic mean, $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$

We know that, $\sigma^2 = \frac{\sum_{i=1}^n x_i^2}{n} - \bar{x}^2$

$$\left(\sum_{i=1}^n x_i\right)^2 = \left(\sum_{i=1}^n x_i^2\right) + \left(2\sum_{i<j}^n x_i x_j\right)$$

$$\therefore \left(\sum_{i=1}^n x_i\right)^2 \geq \left(\sum_{i=1}^n x_i^2\right)$$

$$\Rightarrow (n\bar{x})^2 \geq \left(\sum_{i=1}^n x_i^2\right)$$

$$\Rightarrow n^2 \bar{x}^2 \geq \left(\sum_{i=1}^n x_i^2\right)$$

$$\Rightarrow \frac{n^2 \bar{x}^2}{n} \geq \frac{\left(\sum_{i=1}^n x_i^2\right)}{n}$$

Again,

$$\bar{x}\sqrt{(n-1)} \geq \sigma$$

$$\Rightarrow \sqrt{(n-1)} \geq \frac{\sigma}{\bar{x}}$$

$$\Rightarrow 100\sqrt{(n-1)} \geq \frac{\sigma}{\bar{x}} \times 100$$

$$\Rightarrow 100\sqrt{(n-1)} \geq C.V. \dots\dots\dots(9)$$

[Showed]

$$\begin{aligned} [\because 2\sum_{i<j}^n x_i x_j > 0] &\Rightarrow n\bar{x}^2 - \bar{x}^2 \geq \frac{\left(\sum_{i=1}^n x_i^2\right)}{n} - \bar{x}^2 \\ &\Rightarrow \bar{x}^2 (n-1) \geq \sigma^2 \\ &\boxed{\therefore \bar{x}\sqrt{(n-1)} \geq \sigma} \dots\dots\dots(8) \end{aligned}$$

[Showed]

Theorem: Find mean, variance and standard deviation of first n natural numbers.

Proof: Let, x_i be the i^{th} natural number, i.e. $x_i = 1, 2, 3, 4, \dots, n$

$$\begin{aligned}
\therefore \text{Mean} &= \frac{\sum_{i=1}^n x_i}{n} \\
&= \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} \\
&= \frac{1 + 2 + 3 + \dots + n}{n} \\
&= \frac{\frac{n(n+1)}{2}}{n} \\
\therefore \text{mean} &= \frac{(n+1)}{2} \dots\dots\dots(12)
\end{aligned}$$

Now, we know variance,

$$\begin{aligned}
\sigma^2 &= \frac{\sum_{i=1}^n x_i^2}{n} - \bar{x}^2 \\
\Rightarrow \sigma^2 &= \left(\frac{\sum_{i=1}^n x_i^2}{n} \right) - \left(\frac{\sum_{i=1}^n x_i}{n} \right)^2 \\
\Rightarrow \sigma^2 &= \left(\frac{1^2 + 2^2 + 3^2 + \dots + n^2}{n} \right) - \left(\frac{n+1}{2} \right)^2 \\
&= \frac{\frac{n(n+1)(2n+1)}{6}}{n} - \left(\frac{n+1}{2} \right)^2 \\
\Rightarrow \sigma^2 &= \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} \\
\Rightarrow \sigma^2 &= \frac{(n+1)}{2} \left(\frac{2n+1}{3} - \frac{n+1}{2} \right) \\
\Rightarrow \sigma^2 &= \frac{(n+1)}{2} \left(\frac{4n+2-3n-3}{6} \right) \\
\Rightarrow \sigma^2 &= \left(\frac{n+1}{2} \right) \left(\frac{n-1}{6} \right) \\
\therefore \sigma^2 &= \frac{n^2-1}{12} \dots\dots\dots(13)
\end{aligned}$$

Again, standard deviation,

$$\sigma = \sqrt{\frac{n^2 - 1}{12}} \dots\dots\dots(14)$$

Skewness:

Skewness means “lack of symmetry” i.e. departed from symmetry of a distribution.

A distribution is said to be symmetric if –

- a) If mean, median and mode give different values.
- b) If Q_1 and Q_2 are not equidistance from the median.

Skewness are two types –

- i) **Positive skewness:** If the curve has a long tail in the positive side of the graph (Graph 3, here, $\bar{x} > M_e > M_o$).
- ii) **Negative skewness:** If the curve has a long tail in the negative side of the graph (Graph 2, here $\bar{x} < M_e < M_o$).

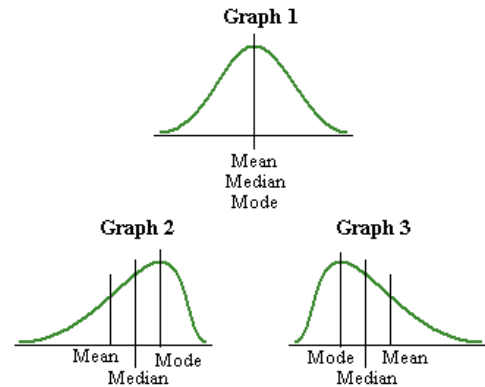


Figure: Skewness

Coefficient of skewness:

Co-efficient of skewness is denoted by β_1 where, $\beta_1 = \frac{\mu_3^2}{\mu_2^3} \cdot \gamma_1 = \sqrt{\beta_1}$.

- ⌚ If $\beta_1 = 0$ and $\gamma_1 = 0$, the distribution is symmetric.
- ⌚ If $\beta_1 > 0$ and $\gamma_1 > 0$, the distribution is positive skewed.
- ⌚ If $\beta_1 < 0$ and $\gamma_1 < 0$, the distribution is negative skewed.

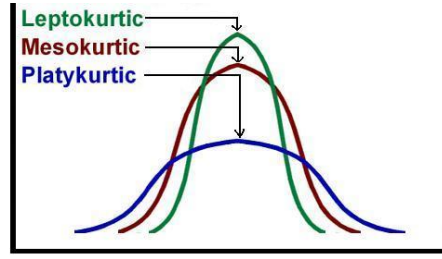
Kurtosis:

The degree of peakness or flatness of a distribution relative to a symmetrical distribution is called kurtosis.

Types of kurtosis

- a) Leptokurtic: If the curve is more peaked than the normal curve.
- b) Platykurtic: If the curve is more flat topped than the normal curve.

c) Mesokurtic: The normal curve itself.



Kurtosis

Coefficient of kurtosis:

Co-efficient of kurtosis is denoted by β_2 and $\beta_2 = \frac{\mu_4}{\mu_2^2}, \gamma_2 = \beta_2 - 3$

⌚ If $\beta_2 = 3$ the distribution is mesokurtic.

⌚ If $\beta_2 > 3$ the distribution is leptokurtic.

⌚ If $\beta_2 < 3$ the distribution is platykurtic.

Examples:

No.	Coefficients	Comments
1.	$\beta_1 = 0$ & $\beta_2 = 1.5$	Symmetric and platykurtic curve.
2.	$\beta_1 = 1$ & $\beta_2 = 3$	Positive skewed and mesokurtic curve.
3.	$\beta_1 = -1.5$ & $\beta_2 = 4.5$	Negative skewed and leptokurtic curve.
4.	$\beta_1 = 0.40$ & $\beta_2 = 3$	Positive skewed and mesokurtic curve.
5.	$\beta_1 = -0.40$ & $\beta_2 = 2$	Negative skewed and platykurtic curve.
6.	$\beta_1 = 0$ & $\beta_2 = 3$	Symmetric and mesokurtic curve.